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Authentic Use of Hattie's Meta-analyses to Frame Educational Research: a Researcher's Reflection

Holley L. Handley[1] · **Nancy B. Hastings**[1]

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Abstract
This article is a formative reflection that walks the reader through the experience of taking an existing body of research and then using one aspect of that analytical framework to conduct a new and original study. This reflection presents an account of that research process. Hattie (2009) has conducted extensive research examining influences on student achievement. This article describes how a researcher took the domains he defined to categorize the influences on student achievement and used them to frame the newly developed investigation. This article reflects on the unique application of Hattie's domains as a lens to examine a K-12 organization. An overview of the study and a summary of the findings are presented to give the reader context. The focus of this article is a detailed reflection on how the research was conceived and formed. The following six questions guide this reflection:

1. Was the experience useful?
2. What was learned?
3. How did this experience facilitate personal growth/development?
4. How was this experience impactful?
5. What would, on reflection, be done differently?
6. Were the research choices well-reasoned?

A thoughtful reflection on the process serves to inform decisions critical to formulate potential additional studies. Reflecting on the past process provides thoughtful insights for future research.

Keywords Reflection · Formative · Lens · Collaboration · Questioning

Introduction

This article presents a personal, formative reflection of using Hattie's domains as a lens through which to view and research student performance at a charter school (Hasting & Handley, 2019). While Hattie's meta-analysis is typically used as a way to view key influences that impact student learning, a further question would be to as if Hattie's domains could be used in a more formative fashion to see, focus, and modify the educational process as it is happening? This is a unique use of Hattie's meta-analysis. In the research, Hattie's domains were used to examine a K-12 organization in a structured, systemic, and systematic way. The intent of this reflection is to convey the experience of envisioning and then apply a novel research model to the reality of a K-12 school located in the panhandle in western Florida. Hattie's meta-analysis was used as a lens, or as (i.e., a set of formative goggles), to clarify the idea of success and performance. The lens of Hattie's domains became a catalyst to the collaboration between this researcher and the practitioner, which attempted to capture the "eureka" or ah-ha! moments that were all part of the overall research process.

The Original Study

The original research was guided by the work of Hattie (2009). Hattie compiled a comprehensive synthesis of the effects of various education influences and interventions affecting student success and achievement. The aim of his research was to "develop an explanatory story about the key

✉ Holley L. Handley
HHandley@uwf.edu

[1] University of West Florida, 11000 University Blvd, Pensacola, FL 32514, USA

influences on student learning it is not to build another "what works" recipe" (Hattie, p. 6) in which he described the need for a barometer for success. Many interventions aimed to improve success may be effective, but Hattie encouraged educators to analyze the degree to which these interventions work. Such analyses could either occur either at the macro-level of an entire school system or in the formative process at the classroom level. Some interventions work better than others. Hattie's meta-analysis indexed the various influences on student achievement according to the degree to which each influence affected student outcomes. Through the analysis of over 800 meta-analyses, Hattie and his team of researchers identified 138 influences on student success. They organized the identified influences into six categories, or domains, which he referred to as the following: (a) home, (b) student, (c) school, (d) teacher, (e) curricula, and (f) teaching.

The research being reflected upon was a single-case study in which Hattie's (2009) domains of teaching and learning were used as the framework to examine a K-12 organization in a structured, systemic, and systematic way (Hasting & Handley, 2019). Stated differently, the school was viewed through the lens of Hattie's domains to identify attributes perceived to be instrumental in student success in a Charter middle school located in western Florida. This study examined best practices contributing to the long-term sustained success of the middle school students in an effort to maintain and replicate high student achievement. The case study addressed only those domains under a school's control: (a) school, (b) teacher, (d) curricula, and (d) teaching. These categories framed the study and provided the lens to examine the factors responsible for student success.

The population of the selected school consisted of a total of 17 teachers and 4 administrators. The criteria chosen for the teachers to participate in the study were the following: participants must be current employees of the middle school, full-time status, employed by the middle school for more than 1 year and teach a core academic subject. Nine of the teachers were core curriculum teachers, teaching either math, science, social studies, or language arts. In addition to the nine teachers, interviews of the four administrators—the principal, guidance counselor, special needs counselor, and staff administrator—provided data from the administrative perspective (Hasting & Handley, 2019).

The data collected in the study were teacher interviews, administrator interviews, and classroom observations. The coding strategy integrated pre-figured codes aligned with Hattie's (2009, 2011, 2015) domains (school, teacher, curricula, and teaching) while allowing for the possibility for emergent codes. Data collection and analysis led to the identification of three over-arching themes, flexibility, personalized learning, and high expectations, as key attributes contributing to student success. Flexibility was evident in the ability of the school to respond quickly to learning opportunities, teachers' willingness to implement innovative teaching strategies, and the administrators' and teachers' commitment to flexible curriculum based on student needs. Individualized instruction was evident as teachers and administrators had the ability to focus on the individual student. Individualized instruction was facilitated by assessing each student's performance and needs and subsequently delivering curricula and instruction based on the assessments and feedback. The third emergent theme was the culture of high expectations. The reputation and expectation of academic excellence were consistently communicated throughout the school from administrators and teachers. Flexibility, individualized instruction, and culture of high expectations flowed from the awareness of student needs and the mission of the educational institution (Hasting & Handley, 2019).

In the work with Hattie's domains used as a tool to structure and view student learning, the researcher had to think similar to that of a player-coach. This researcher had to continually ask how and why this use of Hattie was effective. Since this researcher had experience in many of the roles being studies, it was important to remain aware of which hat was being worn. The "player hat" (former administrator, teacher, counselor) had to be suppressed while focusing on asking the research questions. Former experience gave valuable insight but could not influence how this researcher conducted the study or recorded and analyzed data. This was difficult and had to be consciously on the researcher's mind during the data collection and analysis.

Most of the previous studies were more summative in nature in that they viewed these domains at the end of the educational process. The unique perspective in the study was to illustrate how the domains could also be used as a lens, during a formative review process to guide the development of hands-on questions: "How are we doing and what can we do better?" While the original research presented above was important, the focus of this article is to reflect on the how the task of doing the study had an impact on this researchers' thinking process and development as a researcher. The reflection then becomes a primary influencer on the formative design of future research and practices.

The Reflection

In order to contextualize this reflection, a relevant analogy is in order. One experience shared by many water lovers is the practice of looking into and through the water from different perspectives. The view from above the water, using the naked eye or possibly polarized glasses, is one angle. Another is the view of a swimmer just above the water's surface. The blurry view of opening one's eyes underwater

is a third. While diving, one often uses goggles or a mask to see more clearly. Each of these four views is a lens through which to see a perspective of reality, and each one makes that reality look very different. To truly understand the complexity of the underwater world, you must integrate the images from all of these angles. Likewise, with research, one must draw on a wide range of perspectives and engage in a reflective, cyclical process to expand a formative understanding of reality.

In 2017, this perspective was applied to the research project described above. Hattie's (2009) work on identifying key influences that impact student learning led this researcher to healthy questioning. This researcher wondered if Hattie's meta-analysis could be used as a lens to study a high-performing K-12 organization and ask a couple of questions:

1. How would this change in view provide feedback about student learning?
2. Could Hattie's meta-analysis be as a formative lens to inform the process of creating effective student learning environments?

The subsequent research resulted in the unique application of the domains identified by Hattie, as a research framework to conduct a qualitative, single-case study (Hasting & Handley, 2019). This article reflects on experiences conducting qualitative research. Reflecting on the process can provide valuable insights on the research process itself and shed light on the iterative nature of investigations. Six questions provide a framework or guide for this reflection.

1. Was the experience useful?
2. What was learned?
3. How did this experience facilitate personal growth/development?
4. How was this experience impactful?
5. What would, on reflection, be done differently?
6. Were the research choices well-reasoned?

Often, the value of research is not just the findings or conclusion of that research, but also in the actual process of how the investigation was conceived, and how it proceeded, evolved, and concluded. The thought processes and decision points along the way can be just as important and informative.

Was the Experience Useful?

The entire process of this research project was very useful to this researcher's development as an investigator. It was exciting to use Hattie's ideas in a way that pushed the limits of his intended use of this work. Hattie's (2009) domains were the product of an enormous amount of data, analysis of that data, and summary of that data into categories he called domains. These data were presented to quantitatively represent how the domains impacted the process of student learning. When reading Hattie's work, the "eureka" moment came when reviewing the domains and the numbers observed by Hattie as something different than observed data. The researcher reflected on other work done in the field of education. Educators often embraced the current popular themes of the day (examples) and applied these themes to the planning and administration of the education process. In the book, *The Shame of a Nation* (Kozol, 2005), Kozol observed that America's education system often adopted new educational practices because they were popular, not because they were empirically proven in a scientific way. A question arose as to how educators could be more influenced by identifying what works in reality, rather than following what is popular at any given time. As Hattie's work was scientific evidence of what is effective and proven by empirical observation. The next question was to consider if educational institutions might be better using the kind of data that Hattie observed to evaluate and measure the effectiveness of specific schools or school systems. A third question was to ask why a more scientific approach could not be utilized to evaluate and improve education that uses real data, real measurements, and real situations.

Hattie's domains were effective as summative measures that they certainly could be very useful to assist with formulating formative assessments and apply then to in-process evaluations.

What Was Learned?

The process of this research also served to teach this researcher to become more self-reflective in research practices. As a rule-follower who prefers established processes, it took considerable time and thought to really embrace Hattie's work from a formative lens. A tension exists between seeking the truth and following scientific protocols, and at the same time being creative and discovering new applications for existing practices. Asking the question "what is true?" should be part of the natural scientific process. The inquisitive part of science should never be completely suppressed by the process. In this reflective process, this researcher learned that the "what if" questions could lead to a more applicable and ultimately more scientific truth: in short, being scientific and creative simultaneously. Scientific in that the data being used was tested for validity and veracity and creativity, in that this was a new use of the science.

Maybe the most profound learning was discovery of the fundamental purpose as a researcher was to learn, grow, and become better about the research process. Senge (2012), in an article published in Leader to Leader, asks an important

question about our education industry are schools learning organizations? Senge goes on to describe schools and educational organizations as systems that should become good at healthy system thinking. In Senge's world, both scientific observation and creative thought prosper simultaneously. These ideas became the catalyst to change the perspective in how Hattie (2009) could be used: as a systemic approach to evaluate how Hattie's (2009) meta-analysis could be used as a lens for viewing schools became one of the ways of offering an improved analysis tool.

According to Senge et al. (2012), learning organizations, schools, systems, classrooms, and even individuals can benefit from both end-process and in-process evaluations. Senge's Fifth Discipline describes the tools of shared vision, mental modeling, team learning, personal mastery, and system's thinking. The in-process, formative uses of these tools effectively merge with using Hattie (2009) as a formative device to assess and adjust while still holding to the scientific process to enhance both formative and summative approaches to designing the learning environment.

How Did This Experience Facilitate Personal Growth/Development?

Part of an educator's personal growth and development is to strive to be a positive influence and to make a difference in the educational system. This idea is true for both the individual level and the macro-, system-wide, level. Senge et al. (2012) work has been particularly influential for the researcher. Senge et al. believe that continued growth and improvement are a system, sub-system, and individual process. He believes that people at every level of a system should strive to become better at what they are doing. It even includes people who, may be a very small part of the educational system, can provide data that could, if used, become valuable input to make choices and improvements to the system. Part of the research process that facilitated personal growth and development was applying a framework in a unique way to provide insights into the performance of a system. Using Hattie's domains as a structure to examine the system, learn about the system, and inform continuous improvements to the system provided the opportunity to learn how to conduct research using a creative approach.

Another water-related analogy is that of a kayak. Using a kayak to go from one destination to another, it is easy to get off-course and end up doing way too much extra paddling. To be successful in the current research, one has to constantly (re)design the work effort and to constantly (re) evaluate the work being done. In process, adjustments had to be made so that the research stayed on course, and taking this to the next step, the information gathered on the current trip provides information for the formation or the "formative design" of the next trip. This same principle can be applied to the process of education. The researcher grew in her ability to reflect, adjust in-process, and design for the future.

This research effort was a growth opportunity in numerous ways. One of these avenues for growth was dealing with the stress of having to think outside of a little box.

In this study, this researcher had to think and function outside of her comfort zone. Creativity was important in sourcing the study since this it was a unique use of Hattie's (2009) domains. This researcher had to think about the domains from the perspective of students, teachers, and administrators, the shareholders in that particular educational system. Developing enhance listening skills was critical during the data collection process. This researcher had to suppress the inclination to lead and teach, but instead listen, record, and do the tedious work of coding data. All of the investigation and analysis allowed growth in the patience and practice of research.

Maybe the best growth opportunity was some of what is called "rabbit chasing." This researcher is naturally curious, and scientific education and training heightened this curiosity. Knowing why things happen is important. Applicable genres or techniques that might expand the knowledge of qualitative research were explored. Many iterations were contemplated for the design of the research study. Three of these design iterations that evolved into the final methodology used to conduct the research are highlighted below.

The first research design considered was an action research study. Human performance technology (HPT) and associated models were contemplated to guide a performance analysis of the school and develop an intervention plan. The HPT model (Van Tiem et al., 2012) offers a proven, methodical process to improve human performance. In this case, the performance that needed guidance was within an academic institution. Utilizing this model could help administrators and other stakeholders to take a holistic approach to conducting performance, gap, and cause analyses. The original idea was to identify growth opportunities for the school. The HPT model would provide both a linear and an iterative progression of events allowing a flexible, but structured, application to the entire process of conducting a performance analysis and developing a plan of intervention for the future. This researcher considered the application of the HPT model in the context of conducting a performance, gap, and cause analysis to develop an intervention and implementation plan for the school as an innovative and applicable approach by applying concepts traditionally applied in the corporate environment. Ultimately, this research approach proved to be logistically not the best fit for the project. Future research opportunities are planned to build upon this original idea of inquiry.

The second stage of revisions occurred when this researcher decided to move to a traditional research methodology. Part of this evolution occurred when this researcher

discovered a variety of theoretical frameworks could be used simultaneously. To evaluate the characteristics contributing to the success of a charter middle school, an expanded view of using different lenses or frameworks need to be considered. Two different frameworks were initially chosen: one designed for administration and one designed for teachers and students. The frameworks chosen needed to be general enough to view the school holistically but specific enough to pinpoint elements that had a direct impact on the student and the student's achievement. For identification of best practices utilized by administrators, utilizing a model developed by Marzano et al. (2005) was initially considered. Based on an analysis of 69 studies and surveys of more than 650 building principals, Marzano et al. (2005) developed a list of 21 leadership responsibilities that have a significant effect on student achievement. These recognized responsibilities were used as the basis for identifying best practices being implemented by the leadership of the charter middle school. For the second framework in this iteration, Hattie's (2009) domains of teaching and learning were considered. This initial research design involved using all six domains. All six domains defined by Hattie were going to be used as the basis for data collection and analysis for factors relating to the teachers and students. As described earlier in this reflection, the final study was based solely on Hattie's domains, and only four of the six domains used. The four domains include those factors that influence student performance that the school can influence. The decision to use these domains as the framework came after many cycles of revisions and research.

Finally, this research led to expanded ideas about qualitative research and exploration of other tools and techniques. The qualitative, case study methodology used in collecting and recording the data used in the research led to searching other techniques and practices as research tools. These included other observation strategies and practices used in qualitative research. Two of these research practices were phenomenology and transcendent observation. Chasing this "rabbit" led to learning more about phenomenology as a valid and useful qualitative research method. Gros (2017) observes that phenomenology has experienced a bit of a revival, especially in the social sciences. In Gros' estimation, one of the fathers of the phenomenological process, Alfred Schutz, laid solid groundwork for using phenomenology in the interpretive social sciences. Schutz used what he called pre-scientific subjectivity to enter the world of phenomenological interpretation, even posing that this subjectivity is interrelated to the phenomenon being observed (Gros). The researcher's experiences, both as an administrator and a teacher, lend themselves to the lived experience of the inquiry. The human experience of observing, conducting the investigation, and reflecting upon the process can be effectively captured through phenomenological research methods. While this venture into the idea of phenomenological and transcendental observation was not used in the original work, it builds a platform for future research, possibly using Hattie's (2009) domains within the phenomenological method of investigation.

The success and applicability of applying Hattie's domains of teaching and learning in this context were fruitful. The original development and application of this framework to both research and analysis were fun and fulfilling experiences. It is motivating to think about further research and expanding this line of research into other schools and organizations. Also, the possibility that someone could use this research to make the classroom or the educational system better is inspiring.

How Was This Experience Impactful?

The research on school performance using the domains defined by Hattie (2009) had a profound impact on this researcher, both professionally and personally. Professionally, this work affirmed the ethic of persistence and hard work as foundations for research. That persistence led the researcher to learn to embrace formative evaluation as an in-process effort. There were many opportunities for procrastination, postponement, and even abandonment of this work effort. This might be true for many who are caught up in a research project. But, in life, very few things take the place of persisting through difficulty. Successfully working through scheduling, data collection, data compilation, and the grunt work of this project has given this researcher even more confidence in being able to subdivide projects. The tedious review and proofing process provided the experience and expertise leading to a new career. Working through the frustrations of things that worked and didn't work relate to the skills used every day, such as analyzing, revising, and multitasking. Learning how to effectively function in spite of feeling under the control of the process and not, personally, in much control of anything was an additional positive outflowing of this research process. This researcher asked those questions that always seem to arise when one is caught up in something bigger than their own little world. Will this project ever be done? Will anyone ever read this research? When will sleep happen? Do friends think that this person has dropped off the face of the earth? What will the kids eat for dinner tonight?

Fine-tuning each step of the process, reviewing, and revising each step had tremendous value. Dividing the overall study into smaller steps or chunks was a valuable lesson learned. Having accomplished what was seemingly an overwhelming task at the beginning from the conceptualization of the study gave this researcher tremendous confidence. Even the iterations and refining were such valuable lessons. Each time a section or process had to be revised, the task

was easier because it was not the first time doing the particular activity. Each step in this research process moved the out-of-control project into a more manageable set of subprojects. One can do new things, even big things, and not be overcome by the complexity or size of the effort. Every revision and every rewrite informed the next and made the next revision and next rewrite that much easier. The entire process of conducting the study, applying a unique framework, taking ownership of the process, reworking, and revising had a tremendous impact on confidence and learning for this researcher.

What Would, on Reflection, Be Done Differently?

Road blocks and obstacles always happen in research, no matter how much you plan, but there are always things one can do to mitigate those obstacles. One of those planning-related obstacles was the tight time window chosen to collect the data. For consistency and personal convenience, interviews were conducted within a 1-week period. Time was blocked for the research and organized in such a way to focus on accurately, efficiently, and expeditiously compiling data. On one hand, this was good, but it created a bit of tension with scheduling interviews. A longer data collection window for more scheduling flexibility would be considered if doing the same research again.

Another consideration for refocusing the research design is to utilize a transcendental phenomenological methodology. While this is a more experiential technique of doing qualitative research, it is appropriate and applicable to the strong people orientation of Hattie's domains. When one thinks about Hattie's meta-analysis, that thought process is usually in the context of numbers, statistics, and percentages. However, a very human-oriented applicability to the use of Hattie's domains exists. While the research brought the idea of phenomenological study into the realm of learning, the original work did not include the resurgence of experiential observation as a research tool. But the researcher might dip her toes in the waters of phenomenology for future research projects.

Were the Research Choices Well-Reasoned?

As stated earlier, the use of Hattie's (2009, 2011, 2015) domains in this research was a unique approach. For this reason, the researcher was particularly attuned to whether the study design was based on sound reasoning. Meta-analysis is a method used to combine and analyze many different research studies in an effort to draw conclusions from multiple sources. Since the researcher's nature is to be analytical, it was natural to gravitate toward a study design based on a sound body of data. The use of Hattie's meta-analysis allowed this research to hinge upon over 1200 influences on student achievement. Experienced personally in the classroom and in the field of education, this researcher has seen many ideas and techniques introduced. Often these are buzzwords or "pop science" ideas unsupported by science, data, and analysis. The reasoning behind the use of Hattie's work related to the use of good data, good analysis, and just good science. The application of good science and solid research need not be restricted to the rear-view mirror of research, but also formative, revising during the process and looking ahead to future applications. Research, and solutions, should inform the next iteration of the process.

Hattie's research examined over 1200 completed meta-analyses about influences on learning, and thus, his research was a "meta-meta-analysis" (Hattie, 2009, p. 3). Hattie's research considered innovations aimed to impact student achievement as discussed in the various meta-analyses to "derive some underlying principles about why some innovations are more successful than others in influencing student achievement" (Hattie, p. 3). The meta-analysis statistical method introduced by Gene Glass in 1976 was the basis for Hattie's analytical technique (Glass, 1976). Prior to that time, schools and administrators relied on literature reviews and manually comparing and contrasting various studies and professional development sessions to decide which methods have the greatest impact on improving schools and overall student achievement (Tyack & Cuban, 1995). According to Tyack and Cuban, despite drastic school reform measures and the abundant research on school performance, the American classroom has not significantly changed in the past 200 years.

Hattie's (2009, 2011, 2015) research was an effort to identify what works best to impact student achievement. As discussed in his original work, most innovations introduced into a classroom are perceived as "working" by educators. Hattie's meta-analysis was an attempt to go a step further and identify what works to the greatest extent. Hattie (2009) explained the concept of everything working by the following statement: "when teachers claim that they are having a positive effect on achievement or when a policy improves achievement this is almost a trivial claim: virtually everything works. One only needs a pulse, and we can improve achievement" (p. 16). Hattie's research ranked the influences on student achievement in relative terms based on the effect size. Hattie determined an effect size of 0.4 to be the average effect of 1 year's schooling. Thus, he referred to this value as the "hinge-point." Interventions above the 0.4 effect size were those influences thought to have a true impact on student achievement.

After reviewing this literature about what others had said about the classroom, the research, and the overall lack of using empirical data to evaluate classroom success, the researcher's analytical nature kicked in, and Hattie's work was reviewed with a more critical eye. The analysis including examining

how Hattie (2009) was presented in a positive light and how his work was questioned by others. Hattie's (2009, 2011, 2015) research has not been without criticisms. Eacott (2017) expressed his opinion that Hattie's work in visible learning (Hattie) had been uncritically followed in an almost cultish fashion; in some cases, to the detriment of Australian school leadership. Eacott's opinions related to the nature of individuals that uncritically followed any accepted path in the education field, through no fault of Hattie. Eacott's primary criticism was related to the behavior of Hattie's followers as people who practice a very data-focused form of scientific management. Hattie himself (2017) responded to the critique of his work in visible learning (Hattie, 2009) as he addressed both the misrepresentation of his work and the labeling of those who follow his work as "cultish." Hattie (2017) clarified that these techniques are often used by naysayers as diversions from the meaning and content of the original work. Hattie stated a "so-what" attitude to those who criticized approaching educational performance and solutions from a science-based perspective. Hattie's position has been supported by writers such as Kozol (2005) who was also critical of popular solutions to education that are not science and fact-based. Kozol applauded efforts, at any stage in the learning process, to steer back toward the goal of effectively educating our students. Kozol's (2005) *Shame of a Nation* clearly highlights failures and a "missing of the mark" at all phases of the education process the classroom domains, formative design, evaluation, education systems, and even national educational policy.

Another examination into the concept and application of meta-analysis was found in the work of Higgins (2016). Rather than take a for-or-against position regarding the work of Hattie (2009), Higgins provided historical background of meta-analysis and suggested the best uses of meta-analyses. Higgins suggested meta-analyses such as that employed by Hattie (2009) were best used to get a holistic view of educational data, utilizing that data, or compilation of that data, to understand and evaluate educational outcomes such as student performance.

In the end, this researcher reasoned that basing classroom achievement on empirical data, rather than what Kozol (2005) would have called popularity, was a better way of finding truth in the myriad of theories, techniques, and practices. This researcher did not want the work to feed into changing or not changing our educational system for unsound and unscientific reasons. When asking the question, "was the choice of this research well-reasoned?" the answer is yes. The research had a solid purpose. The research was needed in a field where classroom actions were sometimes based on little or no data. The research was approached by looking at both the positives of meta-analysis and the negatives observed by some writers. The qualitative methodology was an approach to go beyond statistics and numbers and investigate in a way to add depth beyond sheer mathematics. The research was doable within the time frame and the resources available. And the research could serve as a building block for future use of a more analytical approach to classroom achievement, something Kozol's work said was necessary to get a true picture of the overall success of the American education system.

Conclusions

Reflecting on the experience of using Hattie's domains of learning as a lens through which to view and research student performance has proven to be another valuable and formative exercise in the entire research process. Recounting the experience of envisioning and then applying this unique research framework in the K-12 world inspires future research. Reliving the experience is similar to the process itself. The reflection provides yet another lens through which to view the process and adds a level of clarity to the often murky, cloudy process of research.

The most useful and usable lessons learned from the research project were the following:

The lens through which you choose to view data and research observations is critical to the usability of that research.

At the beginning of this reflection, goggles are used to see more clearly underwater. When you are swimming, it is easy to allow your vision to be blurred by the wind, waves, water clarity, and other factors. You need assistance if you are going to see what is under the water with any clarity. This is how data, information, and knowledge work. More valid data, better information, and more empirical knowledge are the goggles through which researchers clarify what they see and how they interpret observations. Hattie's domains of teaching and learning (2009) were used as the lens because of the amount of data available, the critical analysis of the data, and the structuring of the observations. These data allowed this researcher to see the educational system in an ordered, structured, and analytical way. In an exegetical sense, this researcher was able to see what was there; not trendy terminology, not buzz words, and not what people "felt" worked. Hattie's meta-analysis presented what, in an analytical sense, worked in our current educational system. The beauty of Hattie's lens was that it was effective as a framework to conduct research and the application for use in the formative process became evident.

Data, data analysis, and traditional conclusions related to that data do not preclude other creative uses of that data.

The eureka moments in this project were when the researcher pieced together Kozol's (2005) admonishment of the American education system for not using scientific analysis in their selection of programs/techniques and Hattie's (2009) meta-analysis that offered a huge volume of tested data

for educational decision-makers. While Hattie's data hadn't been used this way before, the researcher asked the question, "why not?" While use of Hattie's domains was seemingly summative in application, why couldn't they be used in formative design and formative application by education practitioners? The researcher's scientific training shouted that data are here to be used by decision-makers and by practitioners. The literature review pointed to a unique use of Hattie's data. The logical thought process led this researcher to try something that was different. If the results of the work were valid and usable, that would be wonderful. If they were not, something of value would be learned. Either way, our collective knowledge base would be expanded.

Research projects are often viewed as endings by the researcher.

Often it is heard: "I have finally finished … Whew, I am glad that is over" is the mantra for most of us when a long, tedious project is concluded. But each project can also be viewed as a beginning. Activities enjoyed by campers, beachgoers, families, and children, include cornhole, volleyball, horseshoes, Frisbee, badminton. Some of these activities require some setup and planning ahead of time. For example, badminton sets include a net, racquets, supports, and shuttlecocks. The setup of all of these things is a process that precludes the ability to have a game. Everyone is glad to get the setup done, but that is not the ending. It just builds the foundation for playing the game, for participating in the activity. Research, at its best, is similar. If research is done well, it offers a foundation and a context for playing the game (including enhancing the chance to win by making formative, in-game, adjustments) and for participating in the activity of conducting research. One research project becomes the jumping-off point for future research and for applying real, accurate knowledge to future research activities. This is the purpose and hope of the work with Hattie's meta-analysis. If, as Kozol (2005) pleads, these data can be used to apply fact-based information to the design of our education system, we can have a positive impact on future student outcomes. And if further study of Hattie's meta-analysis yields new and better information, all the better. It is the researcher's hope that the work has been a foundation for very important activities, the education of people from all walks of life. As Senge et al. (2012) expressed, our education system must, at its center, be a learning organization that grows in knowledge, adapts to environmental changes, and refines its practices so that the mission of education is realized.

Final Thoughts

This reflection began with the image of water and vision; seeing clearly and understanding how our view may be distorted. This researcher pondered the subject of student achievement and the importance of seeing clearly through different lenses. Kozol's (2005) work, *The Shame of a Nation* has the subtitle, "The Restoration of Apartheid Schooling in America." Kozol's observations, accusations, and examples are convincing, convicting, and disturbing, as he describes how the American education system has failed to effectively and equally provide for the education of all people. This research was inspired by contemplating choices made by the educators in the school systems. How has the lack of in-process adjustments, the lack of knowing scientifically derived interventions, and the lack of a clear understanding of where we are in the education process (a distinctly formative issue) contributed to Kozol's observations? Many educators, school boards, and administrators have the very best interest of students in mind as they conduct the business of education across this nation. They look out at their small part of the larger system and see things in a certain way. Kozol says that the result of this sight was the restoration of a very unequal and (I don't use the word lightly) racist system of education. Whether Kozol is right or wrong, his examples show us that our work to make a better system of education must make better choices and must be based on something that gives us a clear, non-distorted view of what is really happening.

Hattie's (2009) meta-analysis provided a sense of confidence in science, data, analysis, and systematic observation. Student, practitioner, and researcher can be brought together using Hattie's domains as a formative bridge. The goal was to look for what clarifies and shows a true picture of reality. The answer can then inform the future.

Declarations

Conflict of Interest The authors declare no competing interests.

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